

5) Rank the functions by order of growth only, for functions without the *.

$(\sqrt{2})^{\lg n}, n^2, n!, (\lg n)!, (\frac{3}{2})^n, n^3, \lg^2 n, \lg(n!), 2^{2^n}, n^{\frac{1}{\lg n}}, \ln \ln n,$
 $n2^n, n^{\lg \lg n}, \ln n, \frac{1}{n}, 2^{\lg n}, (\lg n)^{\lg n}, e^n, 4^{\lg n}, (n+1)!, \sqrt{\lg n}, 2^{2 \lg n}, n, 2^n, n \lg n, 2^{2^{n+1}}$

1

$\ln n$

$\ln \ln n$

n

$n \lg n$

$$\sqrt{2}^{\lg n} = \sqrt{n}$$

$$1^{\lg n} = n^0$$